

Analysis of a popular public biomedical controlled terminology from OBO Foundry

Disclaimer: this analysis does not endorse this modeling style, it illustrates the challenges that it poses for pragmatic knowledge graph applications and discusses ways to mitigate them.

```
@prefix obo: <http://purl.obolibrary.org/obo/> .
@prefix mondo: <http://purl.obolibrary.org/obo/mondo#> .
@prefix owl: <http://www.w3.org/2002/07/owl#> .
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
@prefix skos: <http://www.w3.org/2004/02/skos/core#> .
@prefix oboInOwl: <http://www.geneontology.org/formats/oboInOwl#> .

obo:MONDO_0000009
  a owl:Class ;
  rdfs:label "inherited bleeding disorder, platelet-type" ;
```

individuals could theoretically be the manifestations of the disease in a living thing

```
rdfs:subClassOf obo:MONDO_0002243, obo:MONDO_0002245, obo:MONDO_0003847 ; #
IF: owl:Class'es: "hemorrhagic disease", "blood platelet disease", "hereditary
disease"

owl:equivalentClass [
  a owl:Class ;
  owl:intersectionOf (
    obo:MONDO_0002243 # IF: "hemorrhagic disease"
    [ a owl:Restriction ;
      owl:onProperty obo:RO_0000053 ; # IF: "has characteristic"
      owl:someValuesFrom obo:MONDO_0021152 # IF: "inherited" (it doesn't
have subclasses, it is subClassOf MONDO_0021125 - "disease characteristic")
```

individuals of obo:MONDO_0021152 are hard to imagine - "inherited" < "disease characteristic"

```
]
[ a owl:Restriction ;
  owl:onProperty obo:RO_0004020 ; # IF: "disease has basis in
dysfunction of"
  owl:someValuesFrom obo:CL_0000233 # IF: "platelet"
```

individuals could theoretically be specific diseased platelets in an individual

```
)
]
```

```

] ;

oboInOwl:hasBroadSynonym "blood platelet disease", "platelet disorder" ;
oboInOwl:hasExactSynonym "bleeding disorder, platelet-type" ;
oboInOwl:hasRelatedSynonym "thrombocytopathy" ;

oboInOwl:id "MONDO:0000009" ;

oboInOwl:inSubset mondo:gard_rare, mondo:otar, mondo:rare ;

```

`oboInOwl:inSubset` is an `owl:AnnotationProperty`, while all 3 of its objects are also `owl:AnnotationProperty`'s, and are all subproperties of "subset_property" which is also an `owl:AnnotationProperty`. Basically, they are saying that it is a rare disease and that it's listed in Open Targets resource.

```

oboInOwl:hasDbXref "DOID:2218", "MEDGEN:610", "OMIMPS:231200", "UMLS:C0005818"
; # IF: supposed to be CURIEs (Compact URIs) for the DOID, MEDGEN, OMIM, UMLS
resource IRIs

skos:exactMatch <http://identifiers.org/medgen/610> ; # IF: outdated IRI
skos:exactMatch <http://linkedlifedata.com/resource/umls/id/C0005818> ; # IF:
an individual from Graphwise LLDI rdf:type of 3 owl:Class'es "Disease, Disorder or
Finding", "Conceptual Entity", "Group"
skos:exactMatch <http://purl.obolibrary.org/obo/DOID_2218> ; # IF: an
owl:Class in "Disease Ontology"
skos:exactMatch <https://omim.org/phenotypicSeries/PS231200> ; # IF: this is a
link to a web page, not an IRI; but there is an owl:Class IRI:
<http://purl.bioontology.org/ontology/OMIM/231200> for BERNARD-SOULIER SYNDROME |
BLEEDING DISORDER, PLATELET-TYPE, 1
.

```

Conclusion: in public biomedical "ontologies" OWL notation is used to capture biomedical concepts as classes, connect them into polyhierarchies, connect them to concepts from other "ontologies" via owl axioms, and link together similar/equivalent IRIs from different RDF-esque sources. There is no focus on classes having instances, they are seen as complex controlled terminologies for scientific annotation of research data, e.g. to capture experimental conditions, historically not represented in RDF. `rdfs:subClassOf` is sometimes used loosely, rather as "is part of" or "has broader" than "is a kind of".

Claim: we need disease IRIs that are individuals and not classes to support RDF patterns below.

Patterns (drafts):

```

:_ResearchProject_123 :explores :_MedicalCondition_123 .
:_CourseOfMedicalCondition_123 :isOfCondition :_MedicalCondition_123 ;
:hasParticipant :_LivingThing_123 .

```

Observation/assumption: clients in the spirit of FAIR want to reuse public biomedical "ontologies" and don't want new IRIs.

Observation/assumption: clients want to take advantage of `rdfs:subClassOf` axioms from public biomedical "ontologies" in querying of instance data.

How does UniProt do this?

It implicitly puns `owl:Class`'es of another public biomedical controlled terminology GO as instances of `up:Concept` by using them as objects of `up:classifiedWith owl:ObjectProperty` with `rdfs:range up:Concept`. It leverages inference on class level - `rdfs:subClassOf*` property paths are materialized as `rdfs:subClassOf` triples.

Find all proteins that have some "protein binding" molecular function (GO:0005515) at <https://sparql.uniprot.org/sparql>:

```
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX GO: <http://purl.obolibrary.org/obo/GO_>
PREFIX up: <http://purl.uniprot.org/core/>
select distinct ?p ?go
where {
  ?p a up:Protein;
  up:classifiedWith ?go .
  ?go rdfs:subClassOf GO:0005515 .
}
order by ?p
```

GO:xxx IRIs exist as classes with their definitions and as instances in the same graph.

What if we:

- define MONDO diseases `owl:Class` IRIs as individuals of `gist:Category` (or its subclasses) or `skos:Concept`
- use them as triple objects of any `owl:ObjectProperty`'s that we need for modeling
- have the original MONDO controlled terminology (whose classes will never have instances) loaded into the same graph repo (separate named graph)
- take advantage of the reasoner inferring direct `rdfs:subClassOf` triples out of `rdfs:subClassOf*` property paths ?

For example,

```
select distinct ?research_project where {
  ?research_project :explores ?medical_condition .
  ?medical_condition rdfs:subClassOf :_MedicalCondition_123 .
}
```

returns all research projects that explore any medical conditions that are `:_MedicalCondition_123` or its subclasses in Mondo.

https://apps.semanticarts.com:10035/#/catalogs/training/repositories/Punning_testing/query/s/subClassOf%20%22reasoning%22

Is there anything wrong with this approach?

Alternative solution (in development):

- Define and use public biomedical controlled terminology IRIs as individuals without ever mixing them with their class definitions in the same graph.
- Translate `rdfs:subClassOf` axioms into `gist:hasBroader|skos:broaderTransitive` relationships and use `gist:hasBroader*|skos:broaderTransitive*` in queries to replicates the effects of `rdfs:subClassOf` reasoning.

For example,

```
select distinct ?research_project where {  
  ?research_project :explores ?medical_condition .  
  ?medical_condition gist:hasBroader*|skos:broaderTransitive*  
  :_MedicalCondition_123 .  
}
```

returns all research projects that explore any medical conditions that are `:_MedicalCondition_123` or its subclasses in Mondo.

https://apps.semanticarts.com:10035/#/catalogs/training/repositories/Punning_testing/query/s/hasBroader%20%22reasoning%22

- If any other axioms defined in MONDO that have reasonable meaning for the disease individuals become necessary to take advantage of in querying of instance data, their analogs can be developed at the instance level.